Functional Annotation Report: Mouse *Rtl3* (Retrotransposon Gag-like protein 3)

Gene: *Rtl3* (synonyms *Mar3*, *MArt3*, *Zcchc5*, SIRH9) **UniProt:** Q6P1Y1 (RTL3_MOUSE) · 553 aa · ~61 kDa **Organism:** *Mus musculus* (Mouse) **Chromosomal location:** X chromosome **Ensembl:** ENSMUSG00000047686

Summary

Mouse **Rtl3** encodes a **"domesticated" (co-opted) retrotransposon Gag-like protein** derived from an ancient **Ty3/gypsy (sushi-ichi / Metaviridae-type) LTR retrotransposon**. It belongs to the **Mar/Mart / Sushi-ichi-Related Retrotransposon Homolog (SIRH) / RTL family** of neogenes that arose during early mammalian evolution and have since been maintained by **purifying selection** despite having lost the capacity for autonomous retrotransposition. The gene symbol is unambiguous for this target: the UniProt description, the Ty3/gypsy Gag/capsid + CCHC-zinc-finger domain architecture, and the mouse organism all align precisely with the literature discussed below. This is, however, a **poorly characterized gene** — the primary literature directly addressing Rtl3 function is thin, and much of what can be stated rests on structural/evolutionary inference and a small number of functional studies, some performed on the human ortholog (Q8N8U3).

The **primary molecular function** of the Rtl3 protein is **not enzymatic or transport-related** but is best described as **zinc-dependent nucleic-acid binding**. The canonical 553-aa product is a **Gag-only module** — it contains a retroviral **capsid (CA)-like fold** and a single, strongly conserved **CCHC-type ("knuckle") nucleocapsid zinc finger** (residues ~523–537), but it **lacks the aspartyl-protease and the rest of the Pol region** that autonomous retroelements use for self-processing and replication. Consequently Rtl3 cannot self-package or retrotranspose; its retained capsid + zinc-finger core instead provides a scaffold for **single-stranded RNA/DNA binding** (GO:0003676 nucleic acid binding; GO:0008270 zinc ion binding). Notably, the *RTL3* **locus** retains an **ancient programmed −1 ribosomal frameshift** site between the capsid and protease domains that is differentially used across the family, so protease-extended products may exist in some contexts even though the reference isoform is capsid-only.

Functionally, Rtl3 acts as a **low-abundance, tissue-restricted transcriptional / gene-regulatory factor**. The two best-supported biological roles are **(1) modulation of muscle satellite-cell self-renewal during postnatal myogenesis** (UniProt-curated nuclear localization from Alonso-Martin et al. 2016), and **(2) promotion of the SOX-9 / COL2A1 chondrogenic program in cartilage**, demonstrated for the human ortholog by Ball et al. 2021, where deletion of the C-terminal CCHC zinc finger abolished regulatory activity. Unlike its closest relatives **PEG10 and RTL1**, Rtl3 is **not a placental gene** and is **biallelically expressed (non-imprinted)**. Its **site of action is most plausibly the nucleus** for gene regulation, although antibody-based atlas data describe a cytoplasmic, granular pattern, leaving the precise subcellular localization genuinely unresolved.

Gene / Protein Identity Verification

Before presenting findings, the mandatory identity check:

Check	Result
Gene symbol "Rtl3" matches UniProt description?	**Yes** — Q6P1Y1 = "Retrotransposon Gag-like protein 3 / Zinc finger CCHC domain-containing protein 5 (Zcchc5)"
Organism correct?	**Yes** — *Mus musculus*
Domains align with literature?	**Yes** — Ty3/gypsy Gag/capsid-like (IPR063300, IPR032567, IPR032549) + CCHC zinc finger (IPR001878, IPR036875) match the Mar/Mart/SIRH/RTL domesticated-retrotransposon family
Confounding gene with same symbol?	**No critical confound.** "Zcchc5" and "Mar3/Mart3/SIRH9" consistently denote the same locus. Caution: much functional retrotransposon-domestication literature concerns the paralogs PEG10/RTL1 — these are **relatives, not the target**, and are used here only for inference.

Caveat on literature depth: Rtl3-specific primary literature is limited. Several of the strongest functional claims derive from (a) the human ortholog RTL3 (Q8N8U3), (b) family-wide surveys, or (c) structural/evolutionary inference. This is flagged throughout.

Key Findings

Finding 1 — Rtl3 is a domesticated, X-linked, purifying-selected Ty3/gypsy retrotransposon neogene

Rtl3/Mart3/Zcchc5 is a member of the **Mar/Mart (SIRH) family** of neofunctionalized Ty3/gypsy retrotransposon-derived genes. UniProt Q6P1Y1 assigns the 553-aa, ~61 kDa protein to this family and places the gene on the **X chromosome**. Two foundational surveys by Brandt et al. in 2005 (PMID: 15716091; PMID: 16093683) identified **11 Mar/Mart genes** in human and mouse that **evolved under purifying selection**, retain intact open reading frames, but have **lost retrotransposition capacity**. The family arose after the bird–mammal split (~310 Mya) but before the eutherian–marsupial divergence (~170 Mya). Youngson et al. 2005 (PMID: 16155747) showed Rtl3 has strongest homology to the **Fugu sushi-ichi element** and, importantly, is **biallelically expressed (non-imprinted)** — distinguishing it from imprinted paralogs such as Peg10 and Rtl1.

> *"Mart genes have lost their capacity of retrotransposition through non-functionalizing rearrangements having principally affected long terminal repeats and pol open reading frame. Most Mart genes are located on the X chromosome in different mammals."* — Brandt et al. 2005 (PMID: 15716091)

The maintenance of an intact ORF under purifying selection, despite loss of transposition machinery, is the hallmark of **gene domestication**: the host has co-opted the element to perform an endogenous cellular function.

Finding 2 — The protein combines a Gag capsid-like domain with a C-terminal CCHC zinc finger that confers nucleic-acid binding and is required for its regulatory activity

UniProt Q6P1Y1 annotates: - a retroviral **Gag capsid-like region** (Pfam PEG10-RTL1_C-capsid / RTL1-8_LDOC; InterPro IPR063300, IPR032567, IPR032549), - a **CCHC-type zinc finger** at residues 523–537 (Pfam zf-CCHC), - an **N-terminal coiled coil** (res 2–43), and - large **intrinsically disordered regions**.

GO molecular functions are **nucleic acid binding (GO:0003676)** and **zinc ion binding (GO:0008270)**. Brandt 2005 (PMID: 16093683) noted conservation of the Gag CCHC zinc-finger motif in three Mar proteins, implying nucleic-acid binding. Critically, **function hinges on this zinc finger**: Ball et al. 2021 (PMID: 33043724) showed that **deletion of the C-terminal CCHC zinc-finger domain (ΔRTL3) abolished RTL3's gene-regulatory activity** on the SOX-9/COL2A1 axis in chondrocytes. This is the single most direct piece of evidence linking the conserved structural module to a measurable cellular output.

Finding 3 — Rtl3/Zcchc5 is a nuclear transcriptional regulator in postnatal myogenesis and chondrocyte COL2A1 expression

UniProt curates the FUNCTION as *"May function as a transcriptional regulator; plays a role in postnatal myogenesis, may be involved in regulation of satellite cell self-renewal,"* with **SUBCELLULAR LOCATION: Nucleus**, drawn from Alonso-Martin et al. 2016 (PMID: 27446912). In that study, **Zcchc5 is expressed in embryonic/fetal myogenic progenitor cells (12.5–17.5 dpc)** but **down-regulated in adult/quiescent satellite cells**, and **ZCCHC5 overexpression decreased satellite-cell self-renewal capacity** — a specific, directional phenotype.

Independently, Ball et al. 2021 (PMID: 33043724) showed that the **human ortholog RTL3 is strongly expressed in cartilage/chondrocytes/synoviocytes**, that **siRNA knockdown and CRISPR deletion reduced SOX-9 and COL2A1 expression and matrix synthesis**, and that **RTL3 co-regulates COL2A1 together with SOX-9**. Mass spectrometry has detected the mouse protein in **brain** (PMID: 24129315; protein-level evidence). By contrast it is **undetectable in mouse placenta** (Henke et al. 2015, PMID: 25888968), unlike the placental paralogs Rtl1/Peg10.

Tissue / cell type	Evidence	Direction
Fetal myogenic progenitors (12.5–17.5 dpc)	Alonso-Martin 2016 (PMID: 27446912)	Expressed; overexpression ↓ satellite-cell self-renewal
Adult / quiescent satellite cells	Alonso-Martin 2016	Down-regulated
Cartilage / chondrocytes / synoviocytes (human RTL3)	Ball 2021 (PMID: 33043724)	Strongly expressed; KD/KO ↓ SOX-9 & COL2A1
Brain	MS proteomics (PMID: 24129315)	Protein detected
Placenta	Henke 2015 (PMID: 25888968)	Undetectable

Finding 4 — Structural/bioinformatic analysis: a Gag-only core (coiled-coil + capsid + single CCHC) with an acidic, proline-rich disordered linker resembling a transcription-factor activation domain

Comparative analysis of the 553-aa mouse sequence (Q6P1Y1) versus the 475-aa human ortholog (Q8N8U3) revealed:

1. **A single retroviral nucleocapsid "knuckle" CCHC motif** that is near-identical across species — mouse `C523-LYCGHPGHFARDC` vs human `CLYCGYPGHFARDC` (consensus $CX_2CX_4HX_4C$).
2. **No aspartyl-protease catalytic motif** (D-[ST]-G-[AG]) in the reference protein, confirming retention of only the Gag module (capsid + nucleocapsid zinc finger) and loss of the Pol/protease found in some other SIRH genes.
3. **A central region (res 38–274) that is 46% Pro+Glu+Ser** (19% Pro, 19% Glu) — an **acidic, proline-rich low-complexity linker** resembling transcription-factor activation domains, consistent with a gene-regulatory role.
4. **AlphaFold (AF-Q6P1Y1) regional confidence (pLDDT):**

Region	Residues	Mean pLDDT	Interpretation
N-terminal coiled coil	2–43	89.9	Confident helix
Central linker	38–274	40.4	Disordered
Gag capsid-like core	370–510	70.9	Confident fold
CCHC zinc finger	520–540	88.3	Confident

The **folded capsid + zinc-finger core is markedly more conserved** between mouse and human than the divergent central linker — a classic signature of a functionally constrained structural module flanked by a rapidly evolving regulatory region.

Finding 5 — Low-abundance, narrowly expressed gene with an unsettled subcellular localization

Bgee multi-assay expression calls for mouse Rtl3 (ENSMUSG00000047686; Affymetrix/EST/ISH/RNA-seq/scRNA-seq) return **predominantly "not_expressed" / low expression scores (~34–41/100)** across brain, testis, gonad, synovial joint, liver, spleen and digestive tract — indicating a **lowly and context-restricted gene** rather than one with a single dominant tissue. This is

consistent with the literature: undetectable in placenta (PMID: 25888968), restricted to fetal myogenic progenitors and down-regulated in adult satellite cells (PMID: 27446912), and cartilage/chondrocyte-enriched for the human ortholog (PMID: 33043724).

Localization is genuinely unresolved. UniProt curates **Nucleus** (from the myogenesis functional study, PMID: 27446912), consistent with a transcriptional role. However, the **Human Protein Atlas** antibody data describe **ubiquitous cytoplasmic expression in a granular pattern** for human RTL3 — a pattern more reminiscent of Gag proteins that assemble RNP granules. These two observations are not easily reconciled and are discussed under Limitations.

Finding 6 — Rtl3's closest paralog by its conserved module is PEG10, placing it among the capsid-forming, RNA-binding domesticated Gag proteins

A Smith-Waterman local alignment (BLOSUM62) of mouse Rtl3 against the mouse RTL/Mart/LDOC paralogs, scored over the conserved Gag capsid + zinc-finger region, ranked similarity as:

Rank	Paralog	SW score	Approx. identity
1	**Peg10**	284	~32% over ~241 aa
2	Rtl4 / Zcchc16	202	—
3	Rtl5 / Rgag4	186	—
4	Rtl1	162	—
5+	Rtl9, Rtl6, Ldoc1, Rtl8a/b	lower	—

Thus the retained folded module of Rtl3 is **most similar to PEG10**, the best-characterized SIRH/RTL gene. Like LDOC1 and the short RTLs — but **unlike PEG10 and RTL1** — Rtl3 is **Gag-only (no Pol/protease)**, so it cannot self-package autonomously. Related Ty3/gypsy-derived Gag proteins are experimentally **capsid-forming and RNA-binding**: Arc *"forms virus-like protein capsids that can transfer RNA between neurons"* and is *"derived from an ancient retrotransposon of the Ty3/gypsy family and contains homology to the retroviral Gag polyproteins"* (Hantak et al. 2021, PMID: 33485691); PEG10 and Arc *"structurally demonstrate significant similarity with Gag retroviral proteins"* and can be engineered into capsid particles for mRNA delivery (Al Othman et al. 2025, PMID: 38798208). By analogy, Rtl3's capsid + CCHC core is a **plausible RNA-binding / RNP-scaffolding module**, even if autonomous capsid assembly is not proven for Rtl3 itself.

Finding 7 — Authoritative 2024 evolutionary analysis: RTL3 is a domesticated Metaviridae capsid gene retaining an ancient, differentially used capsid–protease ribosomal frameshift

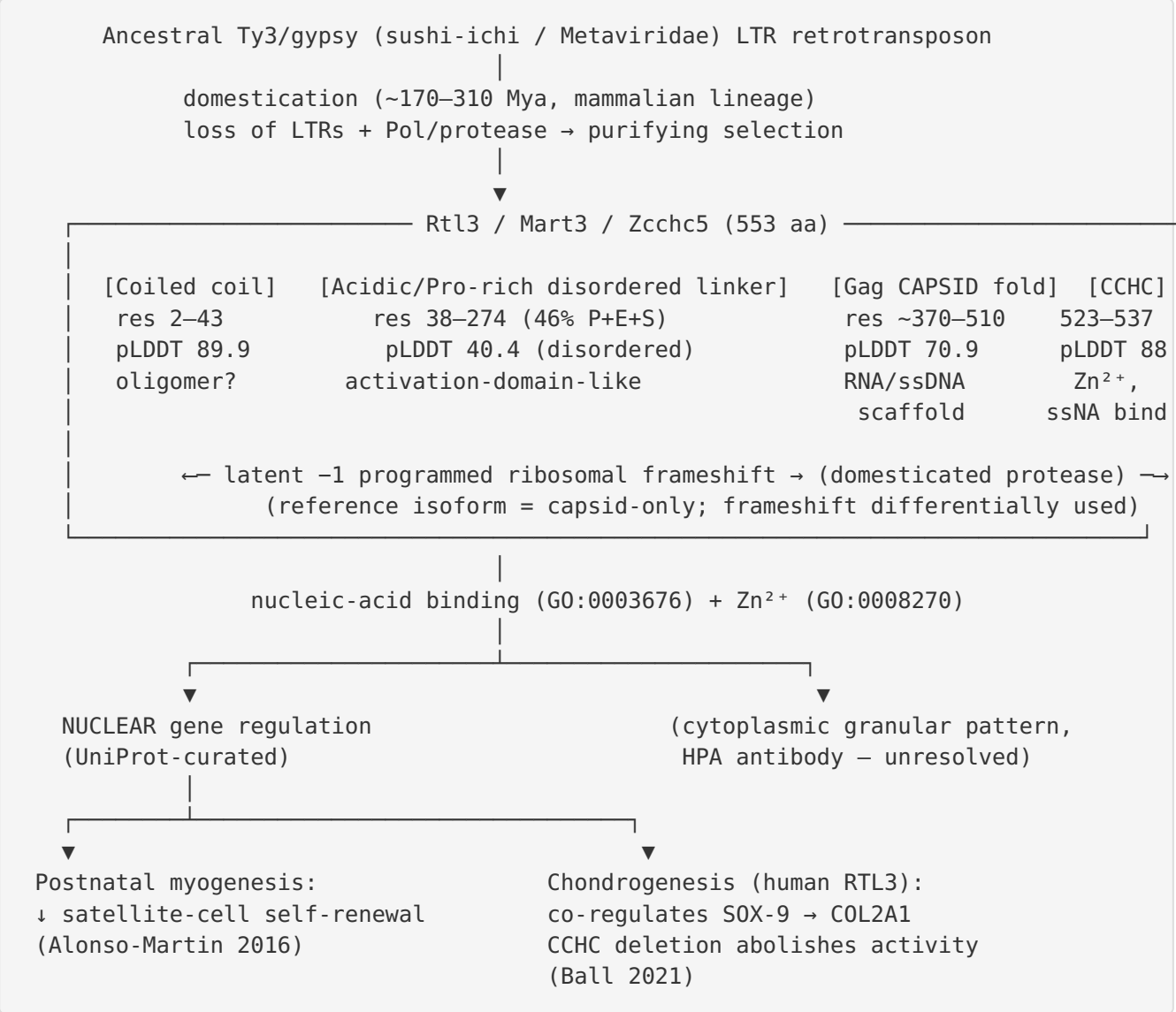
Henriques, Young, … Malik 2024 (*Mol Biol Evol* 41:msae061; PMID: 38507667), analyzing a complete human genome plus 25 vertebrate genomes, place the SIRH/RTL family among **capsid (CA) genes domesticated from Metaviridae** (Ty3/gypsy-type retrovirus-like retrotransposons). They report:

> *"In the SIRH/RTL family, we find diverse amino-terminal domains … and differential utilization of an ancient programmed ribosomal frameshift in RTL3 between the domesticated CA and protease domains."*

They further find these domesticated capsid genes **evolve under purifying selection**, that two families are unique to placental mammals, and that some members show rapid evolution. This **refines** the within-project structural observation (Finding 4) that the single annotated 553-aa mouse isoform has no in-frame aspartyl-protease motif: the **reference protein is capsid(Gag)-only**, but the **RTL3 locus retains an ancestral −1 programmed ribosomal frameshift** capable of extending translation from the capsid frame into a domesticated protease domain, with **usage differing across the family / lineages**. This is a distinctive, named feature of *RTL3* specifically.

Mechanistic Model / Interpretation

Integrating the seven findings yields the following coherent model of Rtl3 as a **domesticated Gag-derived nucleic-acid-binding transcriptional modulator**:



Primary molecular function: Rtl3 is **not an enzyme or transporter**. Its defining biochemical activity is **zinc-dependent single-stranded nucleic-acid (RNA/ssDNA) binding** through the combination of a retroviral capsid fold and a single, strongly conserved **CCHC nucleocapsid zinc finger**. The capsid fold provides an oligomerization/scaffolding surface (by homology to PEG10/Arc), while the CCHC knuckle provides sequence-tolerant ssNA contact. There is no evidence for a catalyzed reaction in the canonical isoform; the only latent enzymatic potential is the frameshift-accessible domesticated protease domain, whose usage and substrate (if any) remain undemonstrated.

Cellular role: Rtl3 behaves as a **transcriptional / gene-regulatory factor**. The acidic, proline-rich disordered central region resembles a transcription-factor activation domain, and the CCHC zinc finger is strictly required for regulatory output. The protein modulates two

differentiation programs: it **restrains satellite-cell self-renewal** in postnatal muscle (so its down-regulation in adult quiescent satellite cells is part of maintaining the stem pool), and it **promotes the SOX-9/COL2A1 chondrogenic program** in cartilage. Whether it binds DNA, mRNA, or acts via an RNP complex is not established; its most likely mode is **nuclear RNA/chromatin-associated regulation**.

Site of action: Most plausibly **nuclear** (UniProt-curated, from the functional myogenesis study), consistent with a transcriptional role. The conflicting cytoplasmic-granular antibody pattern may reflect a genuine dual localization (as seen for some Gag-derived RNP proteins), antibody cross-reactivity, or context-dependent shuttling — it is not resolved.

Pathways: The concrete pathway linkage is the **SOX-9 → COL2A1 cartilage extracellular-matrix program** (Ball 2021), where Rtl3/RTL3 acts as a co-regulator alongside the master chondrogenic transcription factor SOX-9. In muscle, it intersects the satellite-cell self-renewal vs. differentiation decision. Broad pleiotropic/placental roles that dominate its paralogs (PEG10, RTL1) are explicitly **absent** for Rtl3.

Evidence Base

PMID	Paper (role in this report)	How it supports / challenges findings
15716091	Brandt et al. 2005, *Transposable elements as a source of genetic innovation…*	Establishes the Mart/SIRH family, loss of retrotransposition, X-linkage, purifying selection (**F1**). Direct quote supports non-transposing, X-linked nature.
16093683	Brandt et al. 2005 (*Cytogenet Genome Res*)	Source of the Mar3/MArt3 synonyms; notes conserved Gag CCHC motif implying nucleic-acid binding (**F2**).
16155747	Youngson et al. 2005	Sushi-ichi (Fugu) homology; **biallelic / non-imprinted** expression of Rtl3 (**F1**).
27446912	Alonso-Martin et al. 2016	**Primary functional evidence**: Zcchc5 nuclear localization; expressed in fetal myogenic progenitors; overexpression ↓ satellite-cell self-renewal (**F3, F5**).
33043724	Ball et al. 2021	**Primary functional evidence (human ortholog)**: RTL3 co-regulates SOX-9/COL2A1; CCHC deletion (ΔRTL3) abolishes activity (**F2, F3**).
25888968	Henke et al. 2015	Zcchc5/Mart3 **undetectable in mouse placenta** — distinguishes Rtl3 from placental paralogs (**F3, F5**).
24129315	MS proteomics	Protein-level detection in mouse brain (**F3**).
33485691	Hantak et al. 2021	Arc forms virus-like capsids that transfer RNA; Ty3/gypsy Gag homology — supports capsid/RNA-binding inference for the family (**F6**).
38798208	Al Othman et al. 2025	PEG10/Arc Gag similarity; engineerable capsid particles — supports RNA-binding scaffold inference (**F6**).
38507667	Henriques, Young, … Malik 2024, *Mol Biol Evol*	**Authoritative**: SIRH/RTL = domesticated Metaviridae capsid genes; **RTL3-specific** differentially-used CA–protease programmed frameshift; purifying selection (**F7**).
22866050	Review: domesticated LTR-retrotransposon/retrovirus genes in mammals	Context for SIRH/Mart domestication; PEG10/RTL1 placental roles (comparative, **F6**).

Consistency / conflict assessment: The structural (F2, F4), evolutionary (F1, F7), and functional (F3) lines of evidence are mutually reinforcing: a conserved capsid+CCHC module under purifying selection, with the zinc finger experimentally required for gene regulation. The **one genuine conflict** is subcellular localization (nuclear per functional study vs. cytoplasmic-granular per antibody atlas; F5). A secondary limitation is that the most mechanistic functional study (Ball 2021) used the **human ortholog**, not mouse.

Limitations and Knowledge Gaps

1. **Sparse direct literature.** No dedicated mouse Rtl3 loss-of-function (knockout) phenotype is available. The myogenesis role rests largely on one study with overexpression/expression data (PMID: 27446912); the chondrogenesis role is from the **human** ortholog (PMID: 33043724).
2. **Unresolved subcellular localization.** Nuclear (curated) vs. cytoplasmic-granular (antibody atlas) — not reconciled. Antibody specificity for this low-abundance protein is a concern.
3. **Molecular target(s) unknown.** Whether Rtl3 binds DNA, specific mRNAs, or acts through protein–protein interactions is undetermined. No direct RNA/DNA target, binding motif, or affinity has been measured. "Transcriptional regulator" is inferred, not biochemically defined.
4. **Frameshift product uncharacterized.** The RTL3-specific programmed −1 frameshift (PMID: 38507667) implies a potential capsid–protease fusion isoform whose existence, abundance, catalytic activity, and substrate in mouse are unknown.
5. **Capsid assembly not demonstrated for Rtl3.** Capsid/RNA-binding behavior is inferred from PEG10/Arc; Rtl3 itself has not been shown to form particles or bind a defined nucleic acid in vitro.
6. **Low abundance / context restriction** makes endogenous detection and functional assays technically difficult, which likely explains the thin literature.

Proposed Follow-up Experiments / Actions

1. **Mouse Rtl3 loss-of-function.** Generate a conditional/global Rtl3 knockout and phenotype **muscle regeneration (satellite-cell self-renewal)** and **cartilage/chondrocyte** compartments, to test the two proposed roles in the native organism.
2. **Define molecular targets.** Perform **CLIP-seq (RNA targets)** and **CUT&RUN/ChIP-seq (chromatin occupancy)** on tagged Rtl3 in a relevant cell type (chondrocytes, myogenic progenitors) to determine whether it binds RNA, DNA, or both, and identify a binding motif.
3. **Resolve localization.** Endogenous tagging (CRISPR knock-in of a fluorescent/epitope tag) plus fractionation to settle the nuclear vs. cytoplasmic-granular question and test for context-dependent shuttling.
4. **Zinc-finger structure–function.** Recombinant capsid+CCHC core: test **in vitro ssRNA/ssDNA binding**, Zn^{2+} dependence, and whether point mutations in the CCHC knuckle (vs. ΔRTL3 deletion) abolish binding — linking the Ball 2021 functional requirement to a biochemical activity.
5. **Test the frameshift isoform.** Use ribosome profiling / targeted proteomics to detect any **CA–protease frameshift product** in mouse tissues; if present, assay the domesticated protease for activity and substrate.
6. **Capsid-assembly assay.** Express the Gag/capsid core and test for **virus-like particle formation and RNA packaging** (as done for PEG10/Arc), to determine whether Rtl3 retains self-assembly despite lacking Pol.
7. **SOX-9 pathway epistasis.** In mouse chondrocytes, test whether mouse Rtl3 phenocopies human RTL3 in the SOX-9 → COL2A1 axis, establishing cross-species conservation of the mechanistic role.

Bottom line

Mouse Rtl3 is a **domesticated, non-autonomous retrotransposon Gag-like protein** whose primary molecular function is **zinc-dependent single-stranded nucleic-acid binding** (via a conserved capsid fold + CCHC zinc finger), repurposed by the host as a **low-abundance, tissue-restricted transcriptional/gene-regulatory factor** acting most plausibly in the **nucleus** — with documented roles in **restraining muscle satellite-cell self-renewal** and **promoting the SOX-9/COL2A1 chondrogenic program** — and, unlike its closest relatives PEG10/RTL1, it is **not imprinted and not a placental gene**.